

The major limitation of CloudSim is the lack of a graphical user interface (GUI). But despite this, CloudSim is still used in universities and the industry for the simulation of cloud-based algorithms.

Downloading, installing and integrating CloudSim

CloudSim is free and open source software available at <http://www.cloudbus.org/CloudSim/>. It is a code library based on Java. This library can be directly used by integrating with the JDK to compile and execute the code.

For rapid applications development and testing, CloudSim is integrated with Java-based IDEs (Integrated Development Environment) including Eclipse or NetBeans.

Using Eclipse or NetBeans IDE, the CloudSim library can be accessed and the cloud algorithm implemented.

The directory structure of the CloudSim toolkit is given below:

<i>CloudSim/</i>	-- CloudSim root directory
<i>docs/</i>	-- API documentation
<i>examples/</i>	-- Examples
<i>jars/</i>	-- JAR archives
<i>sources/</i>	-- Source code
<i>tests/</i>	-- Unit tests

CloudSim needs to be unpacked for installation. To uninstall CloudSim, the whole CloudSim directory needs to be removed.

There is no need to compile CloudSim source code. The JAR files with the CloudSim package have been provided to compile and run CloudSim applications:

- jars/CloudSim-<CloudSimVersion>.jar*-- contains the CloudSim class files
- jars/CloudSim-<CloudSimVersion>-sources.jar*-- contains the CloudSim source code files
- jars/CloudSim-examples-<CloudSimVersion>.jar*--

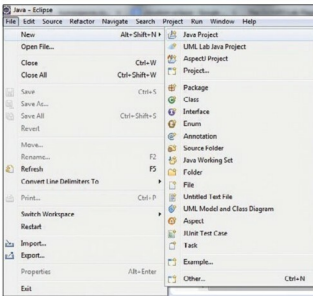


Figure 1: Creating a new Java Project in Eclipse

contains the CloudSim examples class files

- jars/CloudSim-examples-<CloudSimVersion>-sources.jar*-- contains the CloudSim examples source code files

Steps to integrate CloudSim with Eclipse

After installing Eclipse IDE, let's create a new project and integrate CloudSim into it.

- Create a new project in Eclipse.
- This can be done by *File->New->Project->Java Project*

Project

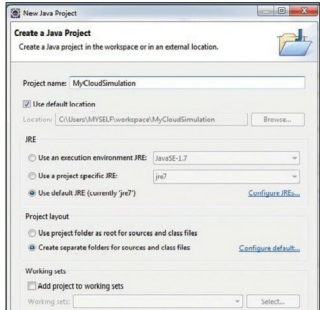


Figure 2: Assigning a name to the Java Project

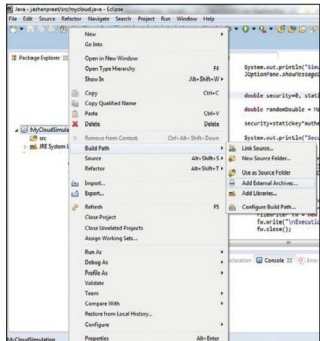


Figure 3: Build path for CloudSim library